

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1830

Little Rock 138 kV
December 23, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1830. The Interconnection Customer proposes a 41.5 MW battery energy storage facility interconnecting at the Little Rock 138 kV substation in the Virginia area. The request was received on July 25, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 560 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,692,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Pahokee to Zephyrhills 138kV
Contingency: Greening 345 kV/138 kV Transformer
Loading: 119.3% (Rating: 231 MVA)
DFAX: 5.2%
Pre-project loading: 118.4%
2. Facility: St Clair to Pahokee 138kV
Contingency: Greening 345/138 kV Autotransformer
Loading: 112.3% (Rating: 225 MVA)
DFAX: 5.2%
Pre-project loading: 111.4%
3. Facility: Ne-Hobbs to Harbor Beach 138kV
Contingency: Mira Vista to Yarnen 345kV line
Loading: 109.3% (Rating: 534 MVA)
DFAX: 10.5%
Pre-project loading: 108.5%
4. Facility: Ne-Hobbs to No Mans Land 138kV
Contingency: Yarnen to Mira Vista 345kV
Loading: 106.5% (Rating: 518 MVA)
DFAX: 10.5%
Pre-project loading: 105.7%
5. Facility: Wheelabrator Portsmouth to Gosport 138kV
Contingency: Chuckatuck to Avra Vly 138kV
Loading: 105.4% (Rating: 371 MVA)
DFAX: 6.3%
Pre-project loading: 104.6%

6. Facility: Bear to Fluvanna P.S 345kV Line
 Contingency: Birchwood to Leavitt 345kV
 Loading: 103.2% (Rating: 2,071 MVA)
 DFAX: 15.1%
 Pre-project loading: 102.9%
7. Facility: Lakeside to Natchez 138kV line
 Contingency: Skaggs Island Junction 2 to Mitchell Dam Substation 345kV line
 Loading: 100.0% (Rating: 625 MVA)
 DFAX: 6.0%
 Pre-project loading: 99.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Zephyrhills to Pahokee 138kV L	Rebuild line with higher-capacity conductor	\$7,852,000	6% (\$468,000)
St Clair to Pahokee 138kV Line	Rebuild line with higher-capacity conductor	\$16,357,000	6% (\$975,000)
Ne-Hobbs to Harbor Beach 138kV	Reconductor line	\$1,731,000	1% (\$16,000)
No Mans Land to Ne-Hobbs 138kV	Reconductor line	\$1,644,000	1% (\$15,000)
Wheelabrator Portsmouth to Gosport 138kV	Reconductor line	\$857,000	2% (\$16,000)
Bear to Fluvanna P.S 345kV	Reconductor line	\$58,243,000	0% (\$198,000)
Lakeside to Natchez 138kV	Reconductor line	\$1,410,000	0% (\$4,000)
	Total allocated to project		\$1,692,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.